

Graduate student Project Description

Stat 842 / CM 7xx Fall 2025

Deadlines:

Deadline for “I cannot find a team member”: Nov 10, 2025

Deadline for sending survey link: Friday 28 Nov 2025, at noon (or earlier)

Deadline for responding to surveys: 3 days after link is sent

Deadline for teams to report names of survey participants: Thursday Dec 4, 2025

Project report deadline: Thursday 11 Dec, 2025 on crowdmark

Designated contact for deadlines: Pamalka Bandera, our Instructional Support Coordinator

Grad students: (from the outline)

The course project is a 2-person team project. You will work on an evaluation for graphs for which you collect data. Specifically, you will design a survey to other students. They will respond.

Undergraduates+Graduates:

You will receive links to surveys in the second half of November. Graduates: Do not fill out your own surveys but do fill out the surveys of other teams.

Sequence:

Project Sequence:

1. Each team of 2 graduate students designs an evaluation and implements the corresponding survey.

Since class email addresses cannot be shared, surveys must be accessible via a link.

2. One team member sends the survey link to the designated contact by the deadline.
3. The link will be distributed to the entire class.
4. Students have 3 days to respond to your survey.
5. After data collection, team members analyze the responses and send the participant list to the designated contact for participation credit.

Note: Sending the survey link earlier will allow you to receive results sooner.

Note on participation credit: Completing surveys for other teams contributes 5% toward your course grade.

Team size:

Each team consists of 2 graduate students. Please, no teams of size 1. You can choose your team member(s) yourself. If you cannot find a team member, please let the designated contact know by the deadline. You will then be paired with somebody else who couldn't find a team member. In your team member disenrolls from the class after the deadline, please let us know as soon as this happens.

Survey tool

Qualtrics is recommended as it is very flexible and freely available to members of the University of Waterloo. <https://uwaterloo.qualtrics.com>

You are welcome to use another tool at your own risk. Make sure you document this in the section "Design".

Report

Each team writes a report 10 pages or less (11pt font or more, single column, 1inch margins on all sides). All team members must contribute to writing and the experiment. On crowdmark, one person hands the assignment in on behalf of the team (make sure to specify your teammate).

You are welcome to include an appendix for completeness (e.g. attach the survey). However, during routine marking we will not look at anything past 10 pages; if there is a problem we might.

What to write about?

Academic writing aims to communicate ideas clearly, not to impress by sounding complicated.

We propose the following structure:

- **Abstract (Summary)**
There is no word limit on the abstract. Something between 75 to 200 words is normal. An abstract is not a cliffhanger: Ideally, the abstract is so well written that the reader does not need to read the whole paper.
- **Introduction**
Introduce the topic of your experiment and explain what your experiment tries to replicate/verify or what gap in the literature your experiment tries to fill. You may cite existing literature if you wish (References may exceed the 10 page limit).
- **Design**
Explain your evaluation. Include a link to your survey. Explain any randomization. Give enough details so we understand what students saw in their survey. In most cases it would be helpful to show 1 (or 2-3?) graphs of what the respondents saw.
- **Results**
Show your data using one or more appropriate graphs. Use an appropriate model to analyze your data. The model may be as simple as a t-test (2 groups), it may be an analysis of variance (multiple groups), or it may be a regression model where you also adjust for one or two covariates. Keep in mind you will get about 70-90 responses; adjusting for 10 covariates is probably an overkill. Avoid unnecessarily complicated regression models (such as neural networks) unless you have a justification.
- **Conclusions**
Succinctly summarise the main finding (s). If applicable, discuss implications for designers of graphs. Discuss limitations of your study.

You may deviate from this structure (with a good reason) but we would not recommend it for most projects.